Coupling between B2.5 and Eirene

X. Bonnin

Bereich Stellaratortheorie, IPP-Teilinstitut Greifswald

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General philosophy

The idea is to replace the neutral fluid model within B2 with the Monte-Carlo kinetic treatment from EIRENE.
The two codes talk to each other via the common block `BRAEIR` and `EIRBRA`. They also create the files:

fort.30 B2 geometry information for EIRENE

fort.31 B2 plasma background for EIRENE

fort.44 EIRENE output for b2plot

B2 computes a plasma background which provides the recycling fluxes to EIRENE. EIRENE then computes the sources and sinks due to neutrals and molecules on this plasma background and passes then back to B2. On the B2 side, the main coupling routine is `eirene_mc.F`. On the EIRENE side, it is `infcop.F`.

Variable settings in B2.5 for coupling

For its array dimensioning, EIRENE uses the parameters set in: `$SOLPSTOP/src/EIRENE/src/common/PARM` as well as some parameters in common with B2, which are in: `$SOLPSTOP/src/include/DIMENSIONS.F`. Additional B2 declarations are found in the directory `$SOLPSTOP/src/Braams/b2/src/common/COUPLE`. You may have the `$SOLPSTOP/src/Braams/b2/B2/B2_include` and `$SOLPSTOP/src/Braams/b2/B2/SOL` directories, whichi contain some obsolete definitions. In that case, make sure that the relevant files from `$SOLPSTOP/src/Braams/b2/B2/B2_include` point to `$SOLPSTOP/src/Braams/b2/src/common/COUPLE` and the ones in `$SOLPSTOP/src/Braams/b2/B2/SOLPS_include` point to `$SOLPSTOP/src/include`. To work in coupled mode, one must do `setenv USE_EIRENE '-DUSE_EIRENE'` and recompile the `b2stbr` and `b2mndr` routines. If working with Eirene version 99 or later, you should also compile (both Eirene and B2) with the preprocessor option `-DNEW_SIZE`.

Within `b2mn.dat`, you should set the following switches:

```
'b2mndt_nstg2'       '10'
'b2mndr_eirene'              '1'
'b2mndr_rescale_neutrals'    '1.0' (on the first coupled run, set to 1.0e-10)
'b2mndr_rescale_neutrals_sources'  '1.0e-10'
'b2stbr_neutrals_namelist'   '1'   (optional)
```

The b2.neutrals.parameters input file

```
&NEUTRALS
NSTRAI=9,
CRCSTRA(1)= 'W', CRCSTRA(2)='E', CRCSTRA(3)='N', CRCSTRA(4)='N', CRCSTRA(5)='N',
RCPOS(1)=    -1, RCPOS(2)   = 59, RCPOS(3)=   24, RCPOS(4)=   24, RCPOS(5)=   24,
RCSTART(1)=   0, RCSTART(2)=  0, RCSTART(3)=  0, RCSTART(4)= 10, RCSTART(5)= 51,
RCEND(1)=    23, RCEND(2)=   23, RCEND(3)=    9, RCEND(4)=   50, RCEND(5)=   58,
CRCSTRA(6)= 'S', CRCSTRA(7)='S', CRCSTRA(8)='V', CRCSTRA(9)='T',
RCPOS(6)=    -1, RCPOS(7)=   -1, RCPOS(8)=    0, RCPOS(9)=    0,
RCSTART(6)=   0, RCSTART(7)= 51, RCSTART(8)=  0, RCSTART(9)=  0,
RCEND(6)=     9, RCEND(7)=   58, RCEND(8)=    0, RCEND(9)=    0,
recyc(1,1)=9*1.00,
userfluxparm(1,1)=0, userfluxparm(1,2)=0,
EIRENE_MOD=1, EIRENE_STEP_DT=100000.0,
volrecinc=1.1, volrecwt=1.0, volrecstart(8)=1.5e21,
chemical_sputter_yield(1)=0.02,
...
chemical_sputter_yield(137)=0.002,
fchar_chemical=1, igass_chemical=2 /
```

The b2.neutrals.parameters variables

Similar structure to the **b2.boundary.parameters** file. Here we have 9 neutral 'strata'. The type of strata is given by CRCSTRA ('V' stands for volume recombination, 'T' for time-dependent), its position by RCPOS, extending from RCSTART to RCEND. The RECYC array is the particle recycling coefficient for each of these sources, i.e. B2 multiplies the EIRENE sources by RECYC. EIRENE is called every EIRENE_MOD B2 iteration, with a time step that is EIRENE_STEP_DT times the B2 time step. The VOLREC... relate to volume recombination sources rescaling. VOLRECSTART is used as an initial upper limit if there is no data from a previous run in **b2.neutrals_save.parameters**. The indices for the CHEMICAL_SPUTTER_YIELD correspond to the wall indices in the EIRENE input file. FCHAR_ and IGASS_CHEMICAL are species indices.

New quantities in b2plot

dab2 Neutral density

dmb2 Molecule density

dib2 Molecular ion density

tab2 Neutral temperature

tmb2 Molecule temperature

tib2 Molecular ion temperature

ahal Atomic H_α emissivity

mhal Molecular H_α emissivity

rfla Radial atomic flux

rflm Radial molecular flux

refa Radial atomic energy flux

refm Radial molecular energy flux

pfla Parallel atomic flux

pflm Parallel molecular flux

pefa Parallel atomic energy flux

pefm Parallel molecular energy flux

f30 Creates the `fort.30` file

f31 Creates the `fort.31` file

brna Particle sources from recycling model

brmo Momentum sources from recycling model

brhe Electron heat sources from recycling model

brhi Ion heat sources from recycling model

Relevant switches and diagnostics

Some useful switches:

b2mndr_eirene Turns on the coupling to Eirene

b2mndr_rescale_neutrals Rescales the fluid neutrals

b2mndr_rescale_neutrals_sources Rescales the fluid neutrals sources

b2stbr_neutrals_namelist Indicates presence of `b2.neutrals.namelist`

eirene_mc_output_style The larger it is, the more output you get

A useful diagnostic to use with EIRENE is `fluxt` $nstra$, where $nstra$ is the number of strata in your run, shows you the evolution of the sources strentgh as computed by EIRENE.

Restarting a run

Under normal circumstances, you can continue a run with the following series of commands from the run directory:

```
cp b2fstate b2fstati
b2run b2mn < input.dat >! run.log &
```

Of course, this means that data fro the last run will be overwritten with the current trun data. If you wish to save the old run data, copy **b2fstate** to some save file, or, even better, build a new directory in which you continue the run. We strongly recommend the latest option whenever input parameters are being changed. In cases where the run was abruptly stopped because of a system failure or the like (i.e. not because the code crashed), you can recover some of the run, provided the switch **b2mndr_savecpu** is not set to zero. If it isn't, the the code will have written a save file called **plasmastate.xxxx**, where **xxxx** is an integer starting at 0000 and ranging up. These are written after **savecpu** seconds of CPU runtime. To recover in that case, do the following:

```
setenv $PLASMASTATE b2mn.exe.dir/plasmastate.xxxx
b2run b2co $PLASMASTATE
```

where **xxxx** is the number of the save file you wish to use. This converts the save file into a **b2fstate** file, and you can then proceed as above.